

Packed in three laminated layer (combine layer of printability, barrier and seal)
Example PET-Al-PE : PET for printing, strength, dimensional stability , aluminium foil for oxygen and moisture barrier and PE for sealant layer
Now used for packing pan masala — Paper - foil -PE,

Inks for printing are typically solvent based or UV - flexo inks
Adhesives are polymers or acrylic or extrudable PE/EVA

Goal is to make biodegradable packet of pan masala or gutka

Readable sachet pf pan masala or gutka

- Outer layer can be made easily biodegradable by paper or other similar materials
- Barrier layer can be made biodegradable by Cellulose based films, chitosan, starch based films or PLA (*but needs to be careful as they may read with pan nasal elements*).
- Seal layer

Elements of Masala or Gutka

Common Ingredients in Pan Masala

Areca nut (supari / betel nut) – sliced or powdered
Catechu (kattha) – extract from acacia wood
Slaked lime (calcium hydroxide / chuna)
Flavoring agents – cardamom, clove, menthol, saffron-type flavors, rose essence, etc.
Sweeteners – sugar, saccharin, other sweeteners
Spices – fennel, coriander, nutmeg, etc.
Humectants / preservatives – depending on manufacturer
Coloring agents – sometimes added

Additional Ingredients Common in Gutka

Gutka generally includes pan masala-type ingredients **plus tobacco**:

Tobacco flakes/powder
Nicotine (naturally present in tobacco)
Sometimes stronger flavorants to mask bitterness

Health-Relevant Chemical Components Often Present

Arecoline and related alkaloids (from areca nut)
Nicotine (from tobacco in gutka)
Tobacco-specific nitrosamines (in tobacco products)
Heavy metals / contaminants may be found in some tested products depending on manufacturing quality
High alkalinity due to lime, which can increase absorption of compounds